# Tech Employment in the Age of AI: Milestone 2 EDA

We Came, We Saw, We Modelled

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DS 4002 | 2026-03-22

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## 1. Setup and Data Loading

```
library(tidyverse)
library(lubridate)
library(scales)
library(forecast)
library(tseries)
library(zoo)
library(ggplot2)
library(patchwork)
library(corrplot)
```

```
df <- read_csv("/Users/jensen.harvey/Desktop/data science/project2/Tech_layoffs.csv")

glimpse(df)
```

```
## Rows: 1,745
## Columns: 18
## $ Nr                          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,~
## $ Company                     <chr> "Tamara Mellon", "HopSkipDrive", "Panda Sq~
## $ Location_HQ                 <chr> "Los Angeles", "Los Angeles", "San Francis~
## $ Region                      <chr> "other", "other", "San Francisco Bay Area"~
## $ USState                     <chr> "California", "California", "California", ~
## $ Country                     <chr> "USA", "USA", "USA", "USA", "USA", "USA", ~
## $ Continent                   <chr> "North America", "North America", "North A~
## $ Laid_Off                    <dbl> 20, 8, 6, 16, 130, 4, 50, 30, 784, 375, 15~
## $ Date_layoffs                <date> 2020-03-12, 2020-03-13, 2020-03-13, 2020-~
## $ Percentage                  <dbl> 40, 10, 75, 100, 22, 20, 50, 50, 98, 15, 1~
## $ Company_Size_before_Layoffs <dbl> 50, 80, 8, 16, 591, 20, 100, 60, 800, 2500~
## $ Company_Size_after_layoffs  <dbl> 30, 72, 2, 0, 461, 16, 50, 30, 16, 2125, 6~
## $ Industry                    <chr> "Retail", "Transportation", "Consumer", "S~
## $ Stage                       <chr> "Series C", "Unknown", "Seed", "Seed", "Se~
## $ Money_Raised_in__mil        <dbl> 90, 45, 1, 6, 79, 4, 17, 7, 120, 1600, 280~
## $ Year                        <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, ~
## $ latitude                    <dbl> 34.05369, 34.05369, 37.77926, 30.27113, 39~
## $ longitude                   <dbl> -118.24277, -118.24277, -122.41933, -97.74~
```

```r
head(df)
```

```
## # A tibble: 6 x 18
##      Nr Company       Location_HQ   Region    USState Country Continent Laid_Off
##   <dbl> <chr>         <chr>         <chr>     <chr>   <chr>   <chr>        <dbl>
## 1     1 Tamara Mellon Los Angeles   other     Califo~ USA     North Am~       20
## 2     2 HopSkipDrive  Los Angeles   other     Califo~ USA     North Am~        8
## 3     3 Panda Squad   San Francisco San Fran~ Califo~ USA     North Am~        6
## 4     4 Help.com      Austin        other     Texas   USA     North Am~       16
## 5     5 Inspirato     Denver        other     Colora~ USA     North Am~      130
## 6     6 Flytedesk     Boulder       other     Colora~ USA     North Am~        4
## # i 10 more variables: Date_layoffs <date>, Percentage <dbl>,
## #   Company_Size_before_Layoffs <dbl>, Company_Size_after_layoffs <dbl>,
## #   Industry <chr>, Stage <chr>, Money_Raised_in__mil <dbl>, Year <dbl>,
## #   latitude <dbl>, longitude <dbl>
```

```r
df <- df %>%
  mutate(
    YearMonth = floor_date(Date_layoffs, "month"),
    Month = month(Date_layoffs, label = TRUE, abbr = TRUE),
    Laid_Off = as.numeric(Laid_Off),
    Percentage = as.numeric(Percentage)
  ) %>%
  filter(!is.na(Date_layoffs), !is.na(Laid_Off))

cat("Date range:", as.character(min(df$Date_layoffs)), "to", as.character(max(df$Date_layoffs)), "\n")
```

```
## Date range: 2020-03-12 to 2025-12-28
```

```r
cat("Total records:", nrow(df), "\n")
```

```
## Total records: 1745
```

```r
cat("Total employees laid off:", comma(sum(df$Laid_Off, na.rm = TRUE)), "\n")
```

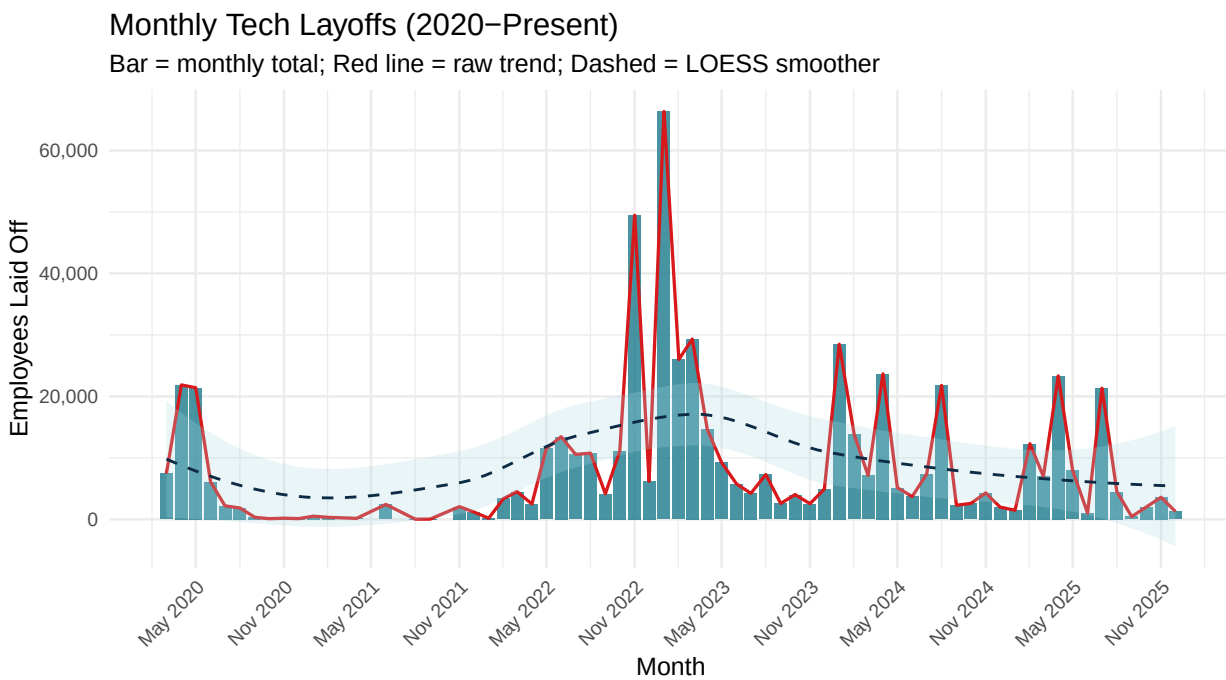
```
## Total employees laid off: 581,705
```

---

## 2. Total Tech Layoffs Over Time (Monthly Aggregation)

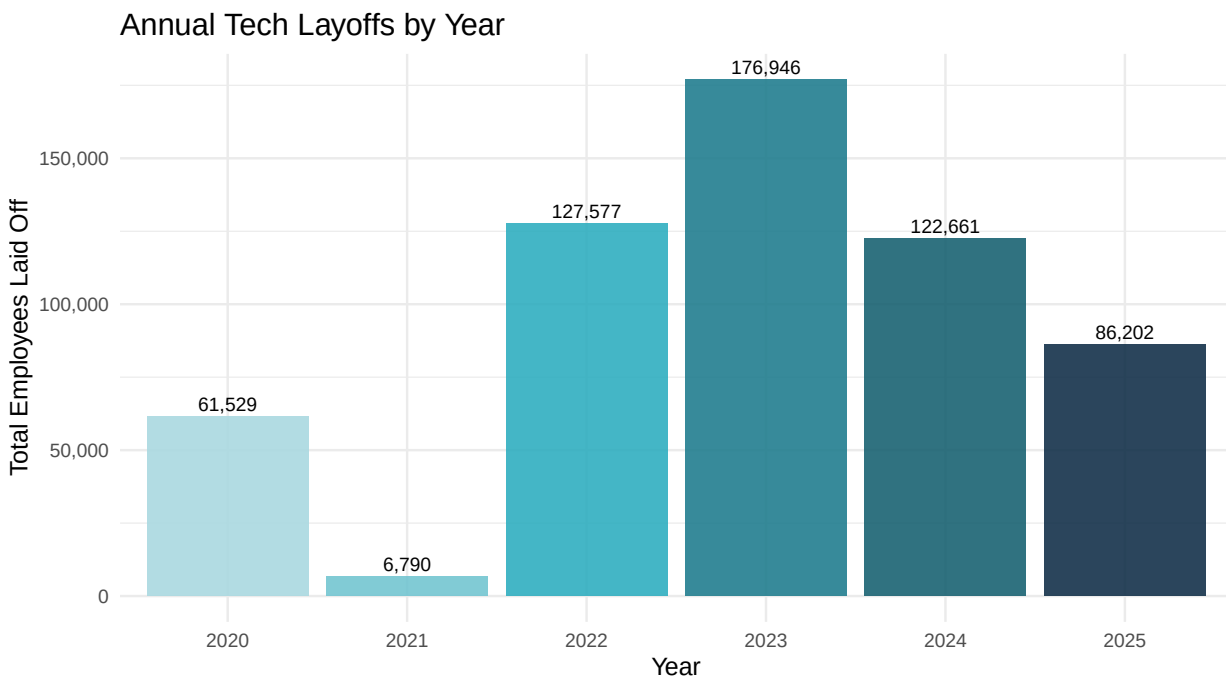
```r
monthly_layoffs <- df %>%
  group_by(YearMonth) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE),
            N_Companies = n_distinct(Company),
            .groups = "drop") %>%
  arrange(YearMonth)

ggplot(monthly_layoffs, aes(x = YearMonth, y = Total_Laid_Off)) +
  geom_col(fill = "#1B7A8C", alpha = 0.80) +
  geom_line(color = "#d7191c", linewidth = 0.8) +
  geom_smooth(method = "loess", se = TRUE, color = "#0D2B45", linetype = "dashed", linewidth = 0.7,
              fill = "#A8D8E0", alpha = 0.25) +
  scale_y_continuous(labels = comma) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  labs(
    title = "Monthly Tech Layoffs (2020-Present)",
    subtitle = "Bar = monthly total; Red line = raw trend; Dashed = LOESS smoother",
    x = "Month", y = "Employees Laid Off"
  ) +
  theme_minimal(base_size = 13) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
annual_layoffs <- df %>%
  group_by(Year) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE),
            N_Companies = n_distinct(Company),
            .groups = "drop")

ggplot(annual_layoffs, aes(x = factor(Year), y = Total_Laid_Off, fill = factor(Year))) +
  geom_col(show.legend = FALSE, alpha = 0.88) +
  geom_text(aes(label = comma(Total_Laid_Off)), vjust = -0.4, size = 3.5) +
  scale_y_continuous(labels = comma) +
  scale_fill_manual(values = c(
    "2020" = "#A8D8E0",
    "2021" = "#6FC4D0",
    "2022" = "#2AACBE",
    "2023" = "#1B7A8C",
    "2024" = "#135E6E",
    "2025" = "#0D2B45"
  )) +
  labs(
    title = "Annual Tech Layoffs by Year",
    x = "Year", y = "Total Employees Laid Off"
  ) +
  theme_minimal(base_size = 13)
```



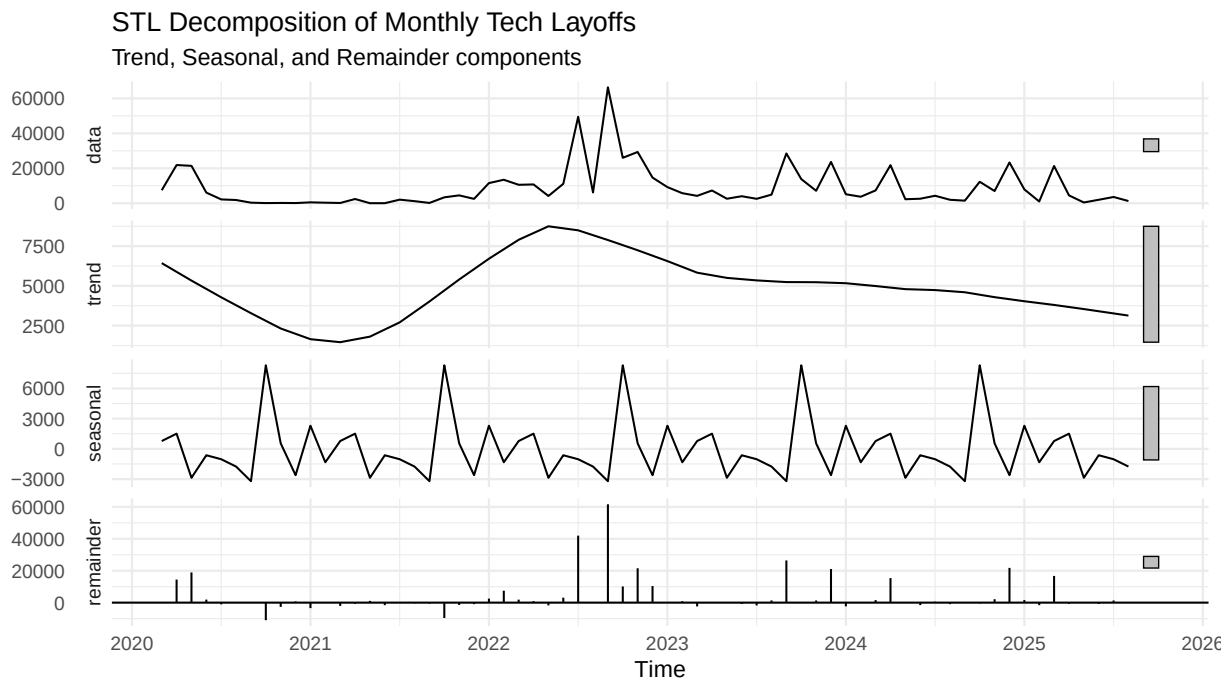

---

## 3. Time Series Decomposition

```
start_year  <- year(min(monthly_layoffs$YearMonth))
start_month <- month(min(monthly_layoffs$YearMonth))

layoffs_ts <- ts(monthly_layoffs$Total_Laid_Off,
                 start = c(start_year, start_month),
                 frequency = 12)

layoffs_stl <- stl(layoffs_ts, s.window = "periodic", robust = TRUE)
autoplot(layoffs_stl) +
  labs(title = "STL Decomposition of Monthly Tech Layoffs",
       subtitle = "Trend, Seasonal, and Remainder components") +
  theme_minimal(base_size = 12)
```




---

## 4. Stationarity Testing (ADF Test) and Differencing

```
adf_raw <- adf.test(layoffs_ts, alternative = "stationary")
cat("ADF Test - Original Series\n")
```

```
## ADF Test - Original Series
```

```
cat("Test Statistic:", round(adf_raw$statistic, 3), "\n")
```

```
## Test Statistic: -2.945
```

```
cat("p-value:", round(adf_raw$p.value, 4), "\n")
```

```
## p-value: 0.1918
```

```
cat("Conclusion:", ifelse(adf_raw$p.value < 0.05, "Stationary", "Non-stationary (differencing needed)")
```

```
## Conclusion: Non-stationary (differencing needed)
```

```
layoffs_diff <- diff(layoffs_ts, differences = 1)
adf_diff <- adf.test(layoffs_diff, alternative = "stationary")
cat("ADF Test - After First Differencing\n")
```

```
## ADF Test - After First Differencing
```

```r
cat("Test Statistic:", round(adf_diff$statistic, 3), "\n")
```

```
## Test Statistic: -4.262
```

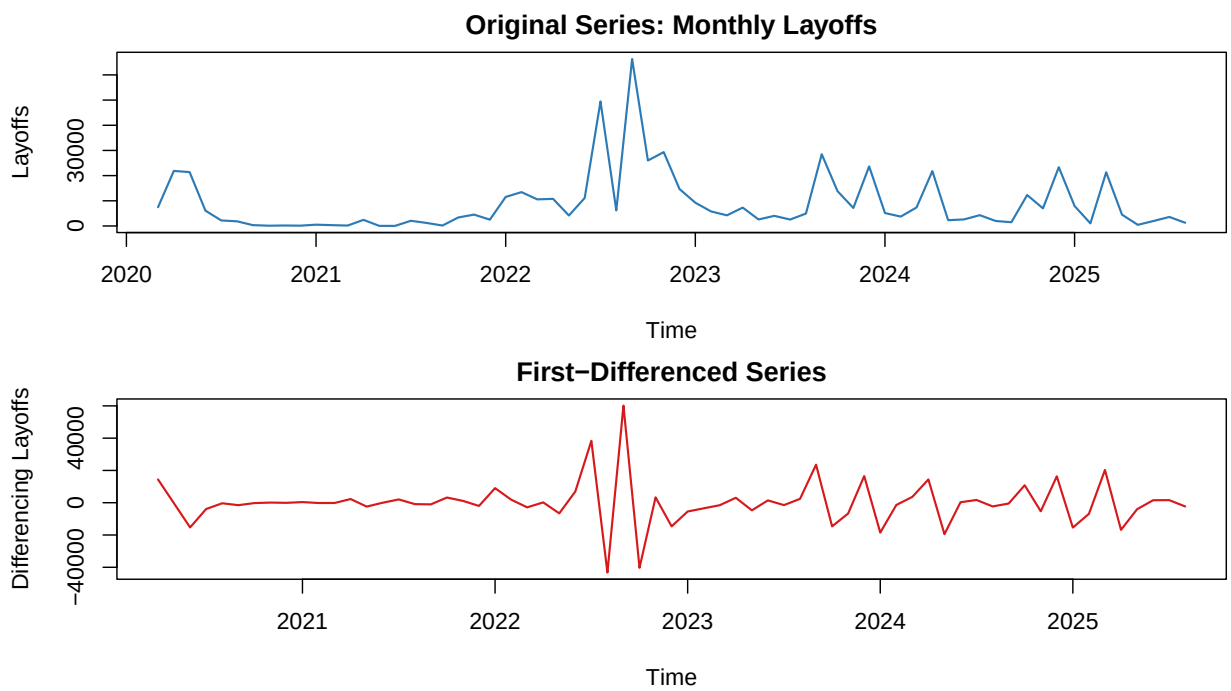
```r
cat("p-value:", round(adf_diff$p.value, 4), "\n")
```

```
## p-value: 0.01
```

```r
cat("Conclusion:", ifelse(adf_diff$p.value < 0.05, "Stationary", "Still non-stationary"), "\n")
```

```
## Conclusion: Stationary
```

```r
par(mfrow = c(2,1), mar = c(4,4,2,1))
plot(layoffs_ts, main = "Original Series: Monthly Layoffs", ylab = "Layoffs", col = "#2c7bb6", lwd = 1.5
plot(layoffs_diff, main = "First-Differenced Series", ylab = "Differencing Layoffs", col = "#d7191c", lw
```




```r
par(mfrow = c(1,1))
```

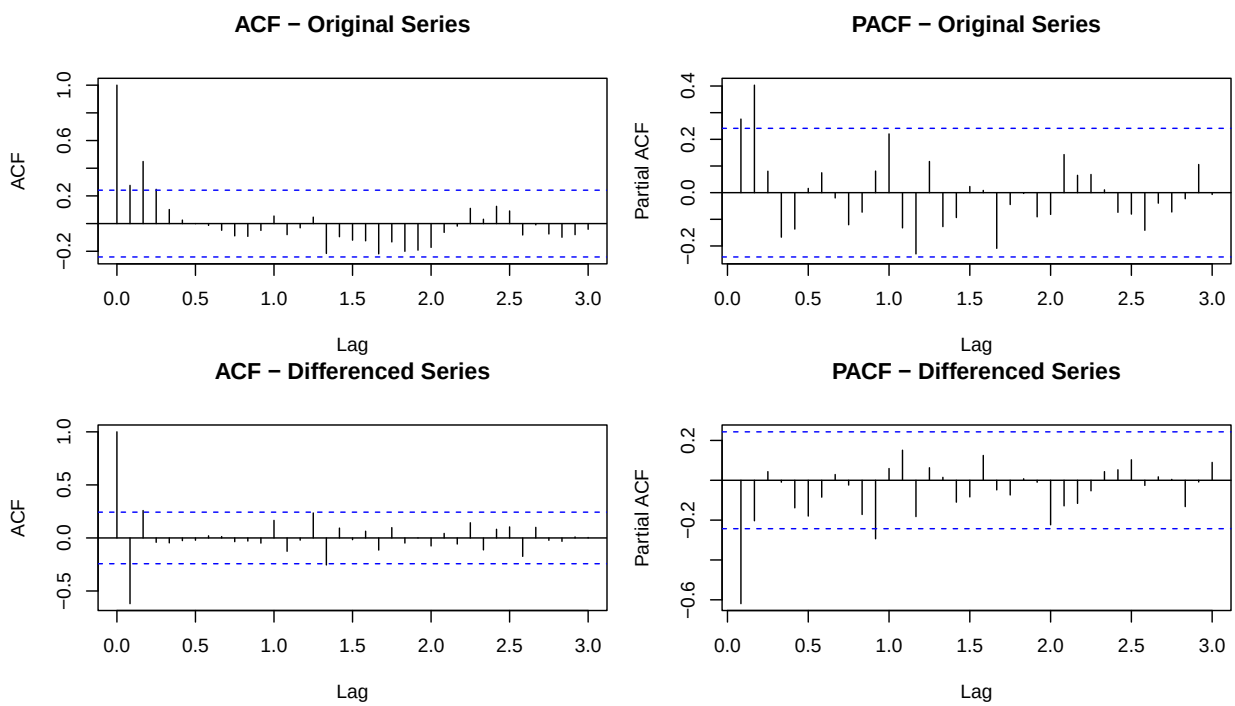
---

## 5. ACF and PACF Plots

```r
par(mfrow = c(2,2), mar = c(4,4,3,1))

acf(layoffs_ts,  lag.max = 36, main = "ACF - Original Series")
pacf(layoffs_ts, lag.max = 36, main = "PACF - Original Series")

acf(layoffs_diff,  lag.max = 36, main = "ACF - Differenced Series")
pacf(layoffs_diff, lag.max = 36, main = "PACF - Differenced Series")
```




```r
par(mfrow = c(1,1))
```

---

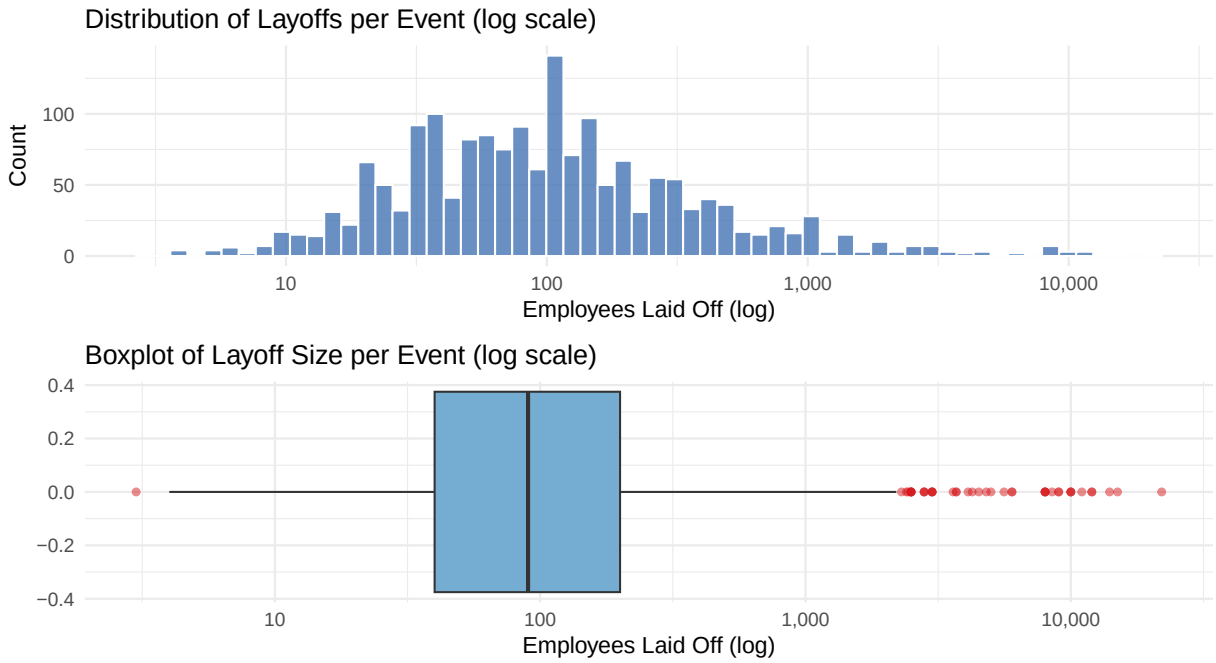
## 6. Distribution of Layoff Events (Company-Level)

```r
p1 <- ggplot(df, aes(x = Laid_Off)) +
  geom_histogram(bins = 60, fill = "#4575b4", color = "white", alpha = 0.8) +
  scale_x_log10(labels = comma) +
  labs(title = "Distribution of Layoffs per Event (log scale)",
       x = "Employees Laid Off (log)", y = "Count") +
  theme_minimal(base_size = 12)

p2 <- ggplot(df, aes(x = Laid_Off)) +
  geom_boxplot(fill = "#74add1", outlier.color = "#d7191c", outlier.alpha = 0.5) +
  scale_x_log10(labels = comma) +
  labs(title = "Boxplot of Layoff Size per Event (log scale)",
       x = "Employees Laid Off (log)") +
```

```
  theme_minimal(base_size = 12)

p1 / p2
```




```
df %>%
  arrange(desc(Laid_Off)) %>%
  select(Company, Date_layoffs, Laid_Off, Industry, Stage) %>%
  head(15) %>%
  rename(`Laid Off` = Laid_Off, Date = Date_layoffs) %>%
  knitr::kable(caption = "Top 15 Largest Single Layoff Events")
```

Table 1: Top 15 Largest Single Layoff Events

| Company | Date | Laid Off | Industry | Stage |
|---|---|---|---|---|
| Intel | 2025-04-23 | 22000 | Hardware | Post-IPO |
| Intel | 2024-08-01 | 15000 | Hardware | Post-IPO |
| Tesla | 2024-04-15 | 14000 | Motor Vehicle Manufacturing | Unknown |
| Google | 2023-01-20 | 12000 | Consumer | Post-IPO |
| TCS | 2025-07-27 | 12000 | IT Services and IT Consulting | NA |
| Meta | 2022-11-09 | 11000 | Consumer | Post-IPO |
| Amazon | 2022-11-16 | 10000 | Retail | Post-IPO |
| Microsoft | 2023-01-18 | 10000 | Software Development | Post-IPO |
| Meta | 2023-03-14 | 10000 | Consumer | Post-IPO |
| Amazon | 2023-03-20 | 9000 | Retail | Post-IPO |
| Microsoft | 2025-07-02 | 9000 | Other | Post-IPO |
| Ericsson | 2023-02-24 | 8500 | Other | Post-IPO |
| Amazon | 2023-01-04 | 8000 | Retail | Post-IPO |

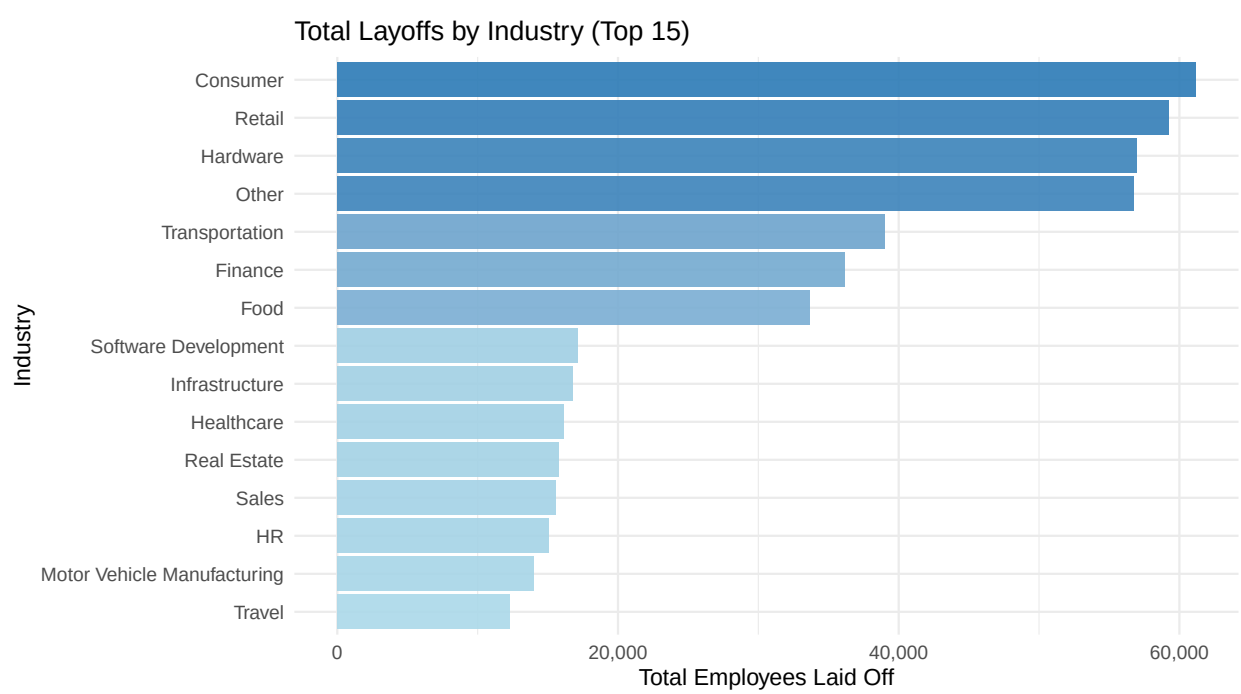
| Company | Date | Laid Off | Industry | Stage |
|---|---|---|---|---|
| SaleSan Francisco Bay Areaorce | 2023-01-04 | 8000 | Sales | Post-IPO |
| Flink | 2023-04-24 | 8000 | Food | Series B |

---

## 7. Layoffs by Industry

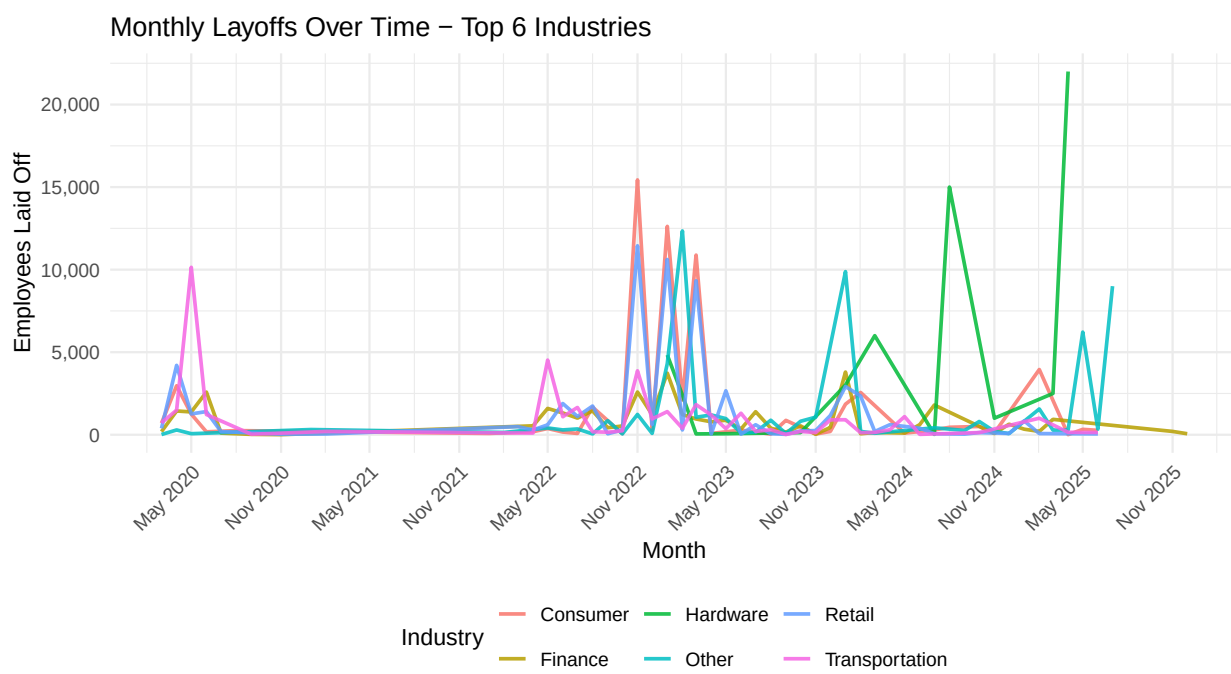
```
industry_summary <- df %>%
  filter(!is.na(Industry)) %>%
  group_by(Industry) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE),
            N_Events = n(),
            .groups = "drop") %>%
  arrange(desc(Total_Laid_Off)) %>%
  head(15)

ggplot(industry_summary, aes(x = reorder(Industry, Total_Laid_Off), y = Total_Laid_Off, fill = Total_La
  geom_col(show.legend = FALSE, alpha = 0.9) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_fill_gradient(low = "#abd9e9", high = "#2c7bb6") +
  labs(
    title = "Total Layoffs by Industry (Top 15)",
    x = "Industry", y = "Total Employees Laid Off"
  ) +
  theme_minimal(base_size = 12)
```



```r
top_industries <- industry_summary$Industry[1:6]

df %>%
  filter(Industry %in% top_industries) %>%
  group_by(YearMonth, Industry) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(x = YearMonth, y = Total_Laid_Off, color = Industry)) +
  geom_line(linewidth = 0.9, alpha = 0.85) +
  scale_y_continuous(labels = comma) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  labs(
    title = "Monthly Layoffs Over Time - Top 6 Industries",
    x = "Month", y = "Employees Laid Off", color = "Industry"
  ) +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom")
```

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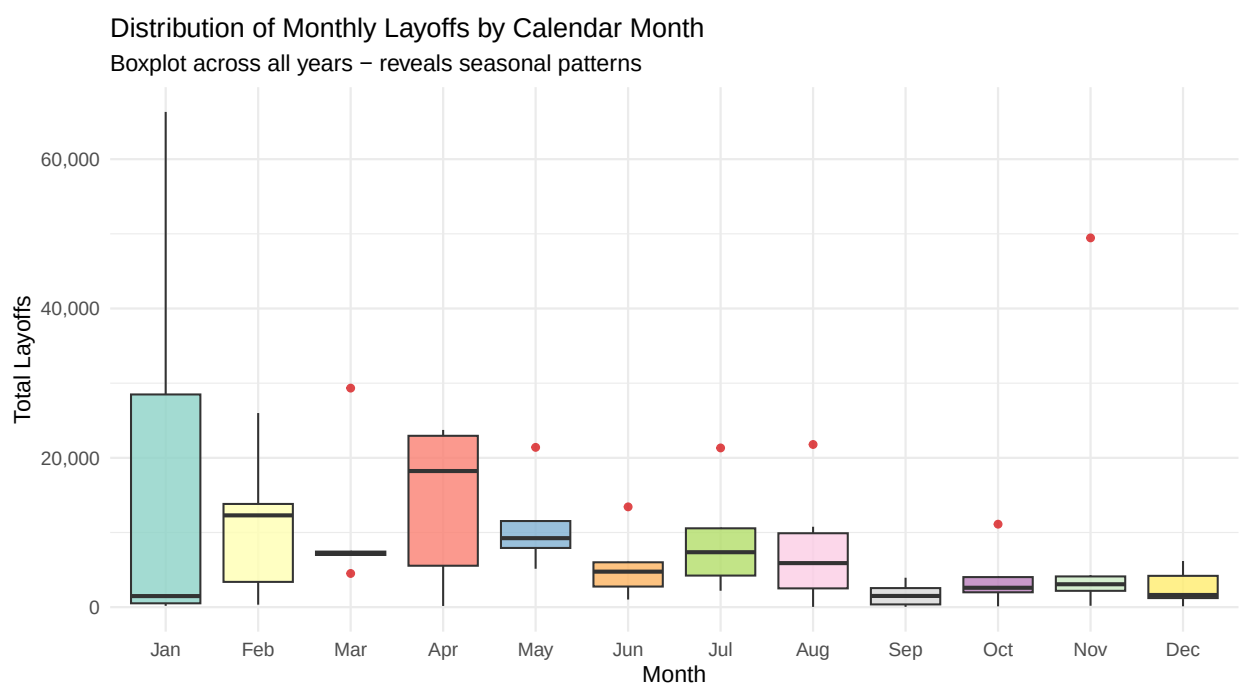
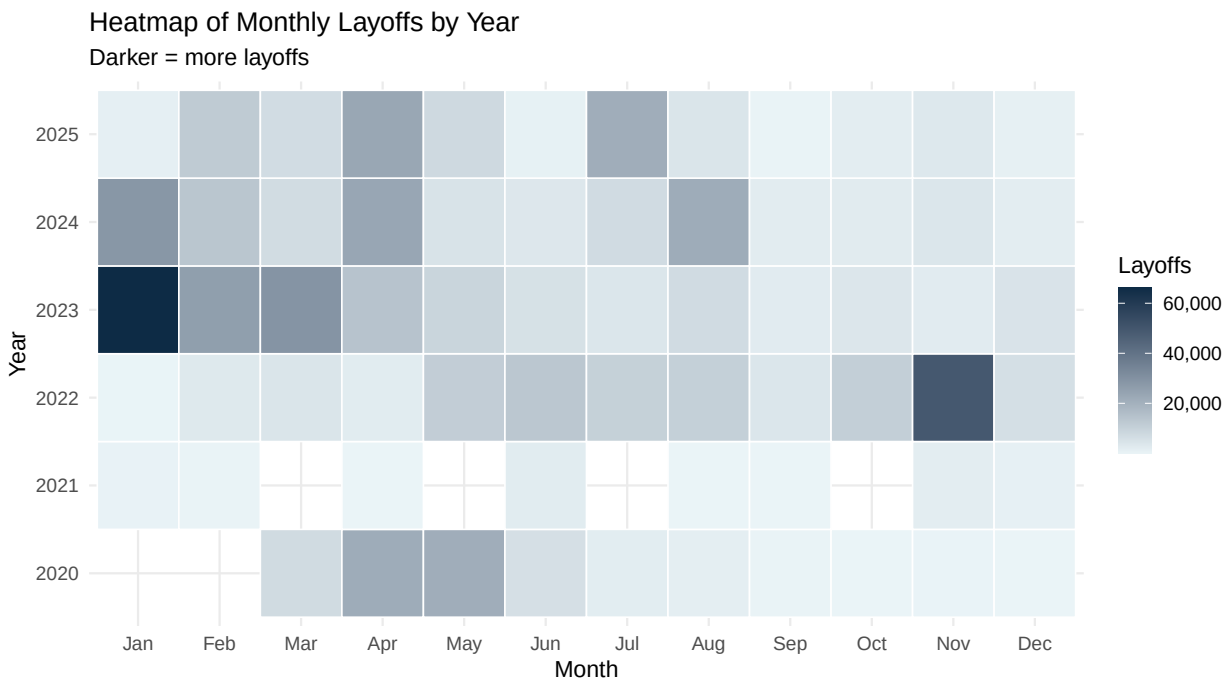
## 8. Seasonality Analysis

```r
df %>%
  group_by(Year, Month) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(x = Month, y = Total_Laid_Off, fill = Month)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.8, outlier.color = "#d7191c") +
```

```r
  scale_y_continuous(labels = comma) +
  scale_fill_brewer(palette = "Set3") +
  labs(
    title = "Distribution of Monthly Layoffs by Calendar Month",
    subtitle = "Boxplot across all years - reveals seasonal patterns",
    x = "Month", y = "Total Layoffs"
  ) +
  theme_minimal(base_size = 12)
```



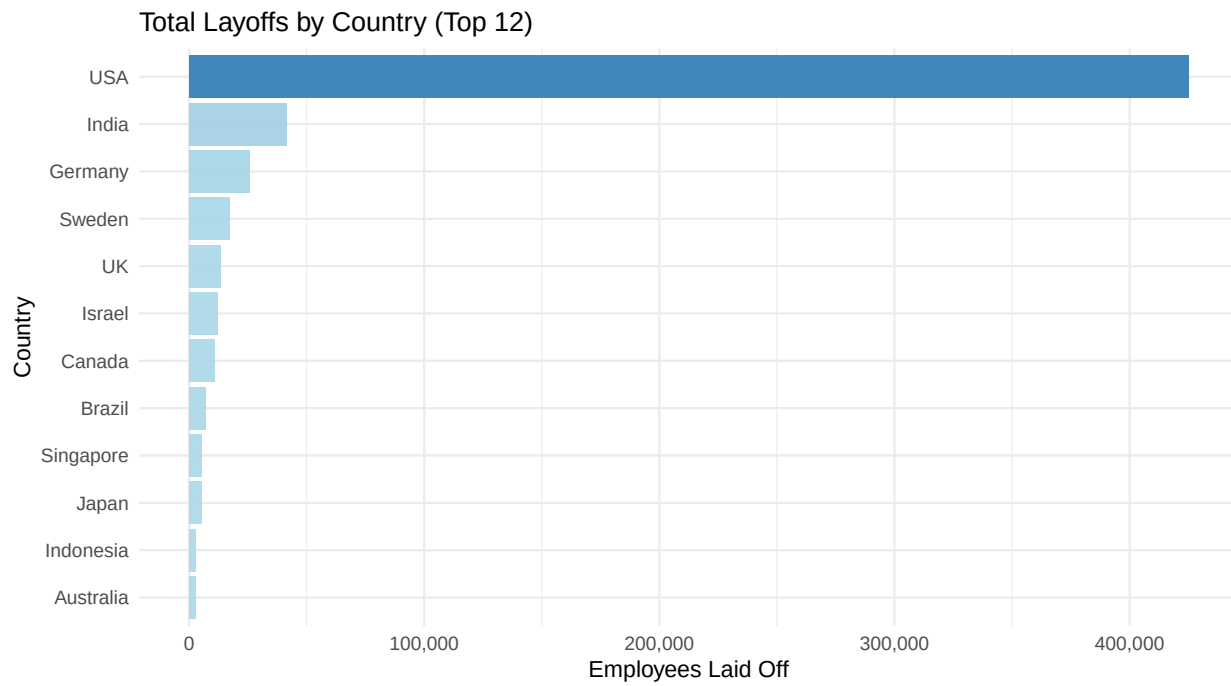

```r
monthly_layoffs %>%
  mutate(Year = year(YearMonth), Month = month(YearMonth, label = TRUE)) %>%
  ggplot(aes(x = Month, y = factor(Year), fill = Total_Laid_Off)) +
  geom_tile(color = "white", linewidth = 0.4) +
  scale_fill_gradient(low = "#EAF4F7", high = "#0D2B45", labels = comma) +
  labs(
    title = "Heatmap of Monthly Layoffs by Year",
    subtitle = "Darker = more layoffs",
    x = "Month", y = "Year", fill = "Layoffs"
  ) +
  theme_minimal(base_size = 12)
```



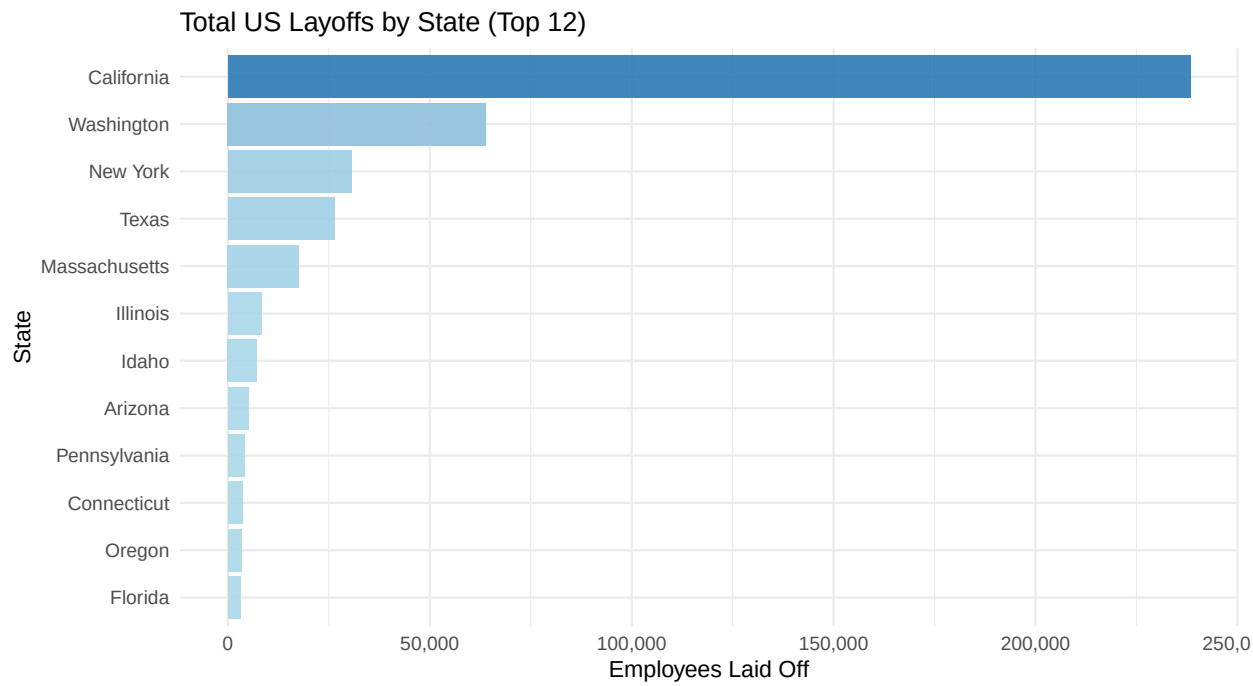

---

## 9. Geographic Distribution of Layoffs

```
df %>%
  filter(!is.na(Country)) %>%
  group_by(Country) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(Total_Laid_Off)) %>%
  head(12) %>%
  ggplot(aes(x = reorder(Country, Total_Laid_Off), y = Total_Laid_Off, fill = Total_Laid_Off)) +
  geom_col(show.legend = FALSE, alpha = 0.9) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_fill_gradient(low = "#abd9e9", high = "#2c7bb6") +
  labs(title = "Total Layoffs by Country (Top 12)",
       x = "Country", y = "Employees Laid Off") +
  theme_minimal(base_size = 12)
```




```r
df %>%
  filter(Country == "USA", !is.na(USState)) %>%
  group_by(USState) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(Total_Laid_Off)) %>%
  head(12) %>%
  ggplot(aes(x = reorder(USState, Total_Laid_Off), y = Total_Laid_Off, fill = Total_Laid_Off)) +
  geom_col(show.legend = FALSE, alpha = 0.9) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_fill_gradient(low = "#abd9e9", high = "#2c7bb6") +
  labs(title = "Total US Layoffs by State (Top 12)",
       x = "State", y = "Employees Laid Off") +
  theme_minimal(base_size = 12)
```

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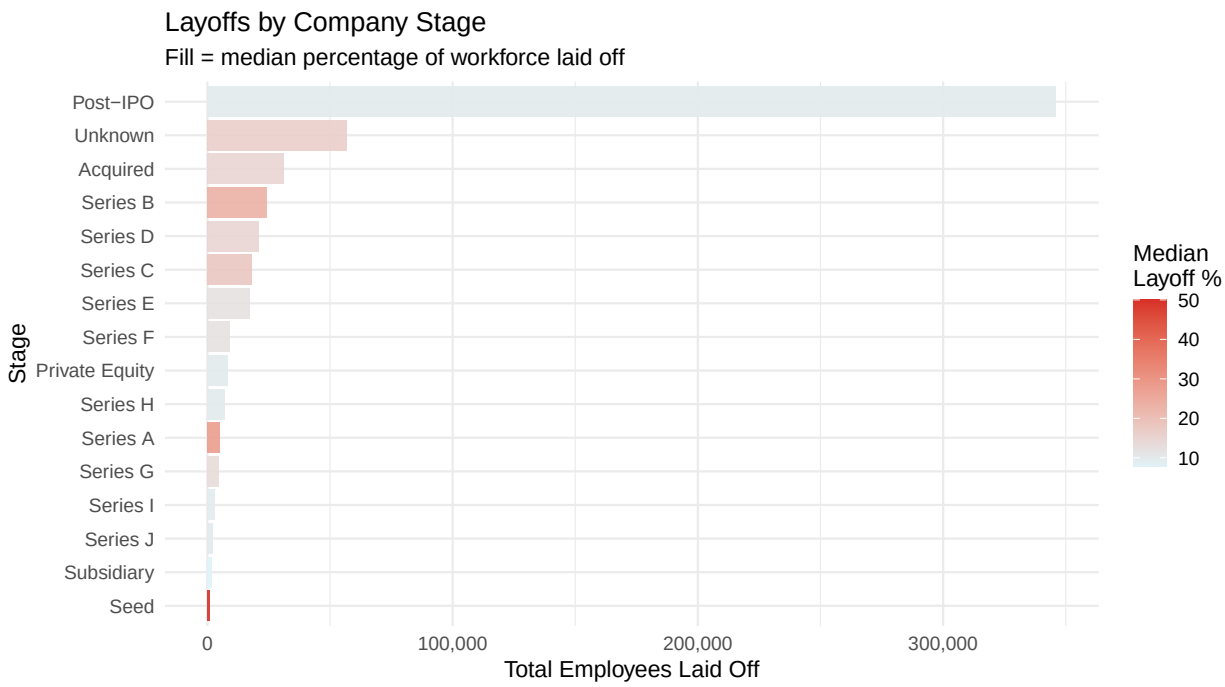
## 10. Layoffs by Company Stage

```
df %>%
  filter(!is.na(Stage)) %>%
  group_by(Stage) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE),
            Median_Pct = median(Percentage, na.rm = TRUE),
            N_Events = n(),
            .groups = "drop") %>%
  arrange(desc(Total_Laid_Off)) %>%
  ggplot(aes(x = reorder(Stage, Total_Laid_Off), y = Total_Laid_Off, fill = Median_Pct)) +
  geom_col(alpha = 0.9) +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_fill_gradient(low = "#e0f3f8", high = "#d73027", name = "Median\nLayoff %") +
  labs(
    title = "Layoffs by Company Stage",
    subtitle = "Fill = median percentage of workforce laid off",
    x = "Stage", y = "Total Employees Laid Off"
  ) +
  theme_minimal(base_size = 12)
```

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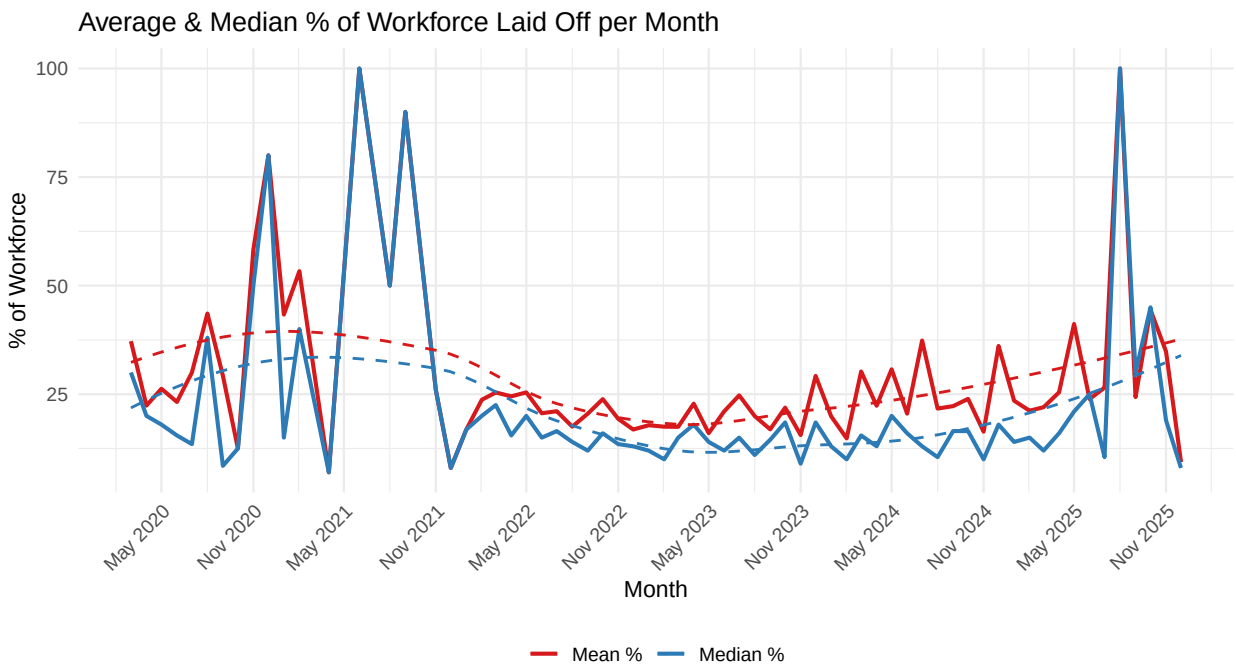
## 11. Percentage of Workforce Laid Off Over Time

```
df %>%
  filter(!is.na(Percentage)) %>%
  group_by(YearMonth) %>%
  summarise(
    Avg_Pct = mean(Percentage, na.rm = TRUE),
    Med_Pct = median(Percentage, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(Avg_Pct, Med_Pct), names_to = "Metric", values_to = "Value") %>%
  ggplot(aes(x = YearMonth, y = Value, color = Metric)) +
  geom_line(linewidth = 1) +
  geom_smooth(method = "loess", se = FALSE, linetype = "dashed", linewidth = 0.7) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  scale_color_manual(values = c("Avg_Pct" = "#d7191c", "Med_Pct" = "#2c7bb6"),
                     labels = c("Mean %", "Median %")) +
  labs(
    title = "Average & Median % of Workforce Laid Off per Month",
    x = "Month", y = "% of Workforce", color = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom")
```


Average & Median % of Workforce Laid Off per Month
100
75
50
25
% of Workforce
May 2020
Nov 2020
May 2021
Nov 2021
May 2022
Nov 2022
May 2023
Nov 2023
May 2024
Nov 2024
May 2025
Nov 2025
Month
Mean %
Median %